Table 1: Measured and calculated current, using $R_{\text{load}} = 390\,\text{k}\Omega$ with 5% tolerance

V_{multi} (V)	I_G (µA)	I_T (µA)
7.44	18.4	19.08 ± 0.96
7.89	20.2	20.2 ± 1.1
9.00	22.0	23.1 ± 1.2
10.13	24.8	26.0 ± 1.3
11.20	27.4	28.7 ± 1.5
12.02	29.2	30.8 ± 1.6

Table 2: Linear fit parameters for $I_T = aI_G + b$

Parameter	Value
a	1.112 ± 0.043
b	$(-1.66 \pm 0.99) \times 10^{-6}$

Table 3: Measured voltage and corrected current

V_{multi} (mV)	I_G (µA)	I_T (µA)
19.6	10.2	9.68 ± 0.86
27.8	13.8	13.68 ± 0.77
43.3	22.2	23.01 ± 0.67
53.2	27.4	28.79 ± 0.70
61.0	30.8	32.57 ± 0.75

Table 4: Calculating galvanometer resistance R_G

Parameter	Value
a	$(5.66 \pm 0.23) \times 10^{-4}$
b	$(-1.6 \pm 1.1) \times 10^{-6}$
$R_G = 1/a$ (kΩ)	1.766 ± 0.071

Table 5: Current comparison between self-made ammeter and multimeter

R_{used} (Ω)	R_{load} (Ω)	V_{PS} (V)	I_G (μA)	I_T (μA)	$I_{\text{tot}} = I_T(1 + R_G/R_{\text{used}})$ (A)	I_{multi} (A)	Error
1.65	150	5.00	28.2	29.68 ± 0.71	0.0318 ± 0.0022	0.032	-0.64%
17	300	1.00	27.8	29.24 ± 0.70	$(3.07 \pm 0.20) \times 10^{-3}$	0.00322	-4.78%
16	150	0.50	26.2	27.46 ± 0.68	$(3.06 \pm 0.21) \times 10^{-3}$	0.00311	-1.67%

Table 6: Voltage comparison between self-made voltmeter and multimeter

R_{used} (kΩ)	R_{load} (Ω)	V_{PS} (V)	I_G (μA)	I_T (μA)	$V_{\text{ab}} = I_T(R_G + R_{\text{used}})$ (V)	V_{multi} (V)	Error
200	150	5.00	23.2	24.13 ± 0.67	4.87 ± 0.28	4.95	-1.66%
50	150	1.50	28.2	29.68 ± 0.71	1.537 ± 0.083	1.497	2.65%

Table 7: Resistance comparison between self-made ohmmeter and multimeter with $V_{PS} = 0.4\,\text{V}$ and $R_{\text{used}} = 6.2\,\text{k}\Omega$

R_{load} (kΩ)	I_G (μA)	I_T (μA)	$R_{\text{load,est}}$ (kΩ)	$R_{\text{load,multi}}$ (kΩ)	Error
0.39	44.2	47.5 ± 1.2	0.42 ± 0.21	0.386	10.08%
39	7.8	7.01 ± 0.92	48.9 ± 7.8	39.2	24.64%
390	0.2	-1.4 ± 1.2	$(-2.8 \pm 2.3) \times 10^2$	390	-173.03%

Table 8: Resistance comparison between self-made ohmmeter and multimeter with $V_{PS} = 2.03\,\text{V}$ and $R_{\text{used}} = 39\,\text{k}\Omega$

R_{load} (kΩ)	I_G (μA)	I_T (μA)	$R_{\text{load,est}}$ (kΩ)	$R_{\text{load,multi}}$ (kΩ)	Error
0.39	49.4	53.2 ± 1.4	-2.49 ± 0.97	0.386	-744.19%
39	24.4	25.46 ± 0.67	39.3 ± 2.9	39.2	0.24%
390	4.2	3.0 ± 1.1	$(6.4 \pm 2.4) \times 10^2$	390	63.33%

Table 9: Resistance comparison between self-made ohmmeter and multimeter with $V_{PS} = 10.09\,\mathrm{V}$ and $R_{\text{used}} = 200\,\mathrm{k\Omega}$

R_{load} (kΩ)	I_G (µA)	I_T (µA)	$R_{\text{load,est}}$ (kΩ)	$R_{\text{load,multi}}$ (kΩ)	Error
0.39	46	49.5 ± 1.2	2.2 ± 5.0	0.386	461.30%
39	38.6	41.24 ± 0.96	42.8 ± 6.1	39.2	9.28%
390	15.6	15.68 ± 0.73	442 ± 37	390	13.25%